STATISTICS & MACHINE LEARNING · LESSON 24 OF 28

Time-Series Forecasting: Decomposition, Baselines, ETS, ARIMA and ML with Lag Features

Forecast demand properly. Components of a time series (trend, seasonality, noise), decomposition with STL, stationarity and autocorrelation (ADF, ACF/PACF), honest baselines (naive, seasonal naive, moving average), exponential smoothing (Holt-Winters), SARIMA, machine learning with lag and calendar features, backtesting with forecast metrics (MAE, RMSE, MAPE, WAPE), prediction intervals, and a guide to Prophet, statsforecast and foundation models.

● ML track (added) · Power BI forecasting ✦ statsmodels 0.1x · LightGBM 4 Advanced ⏱ 130 min

By the end of this lesson you can

- Decompose a series into trend, seasonality and remainder and test stationarity
- Read ACF and PACF plots and difference a series
- Build and evaluate naive, seasonal naive, Holt-Winters and SARIMA forecasts
- Engineer lag, rolling and calendar features and forecast with gradient boosting
- Backtest with a time-ordered holdout, compare models with WAPE/MAPE and report intervals

Datasets used

`daily_sales.csv`

The data

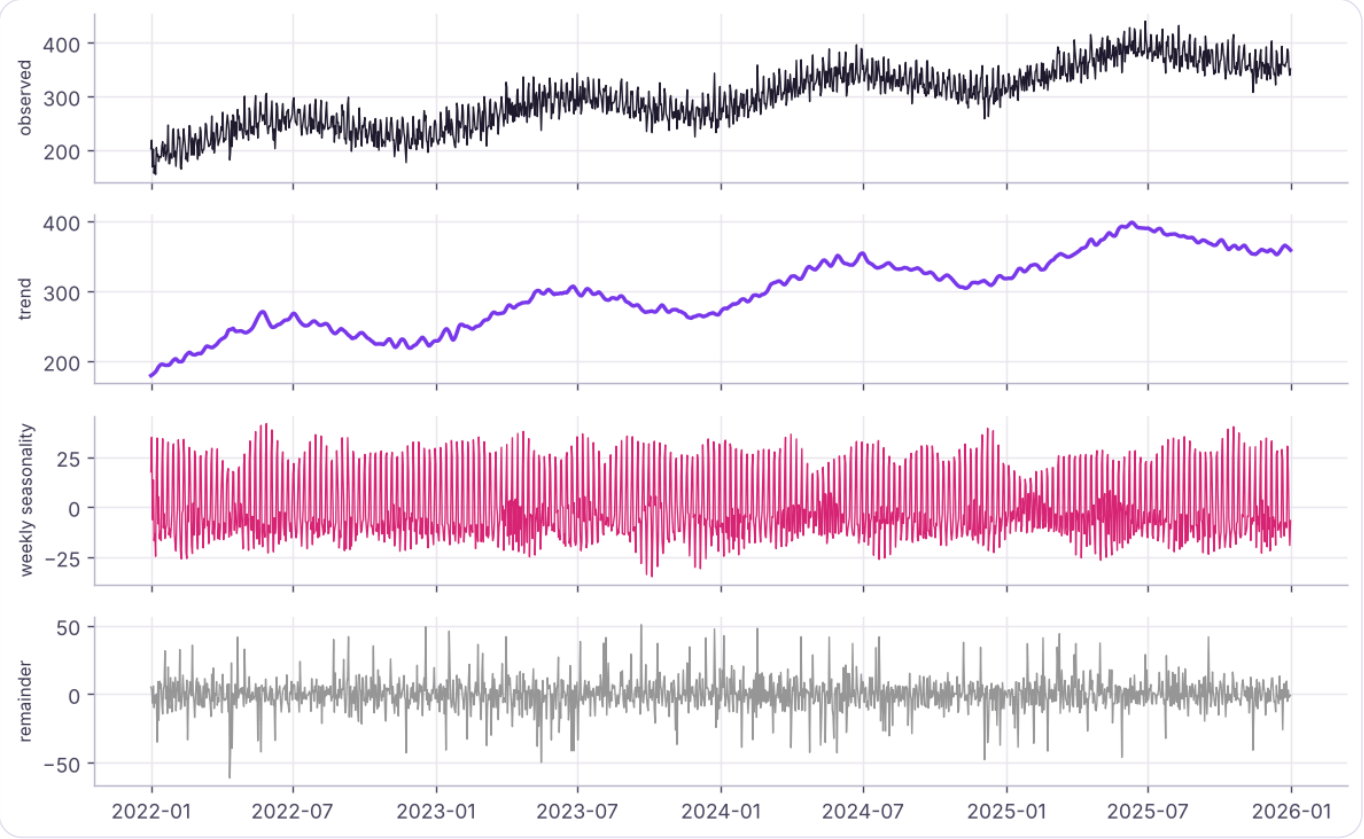
Python

```
import pandas as pd, numpy as np, matplotlib.pyplot as plt, warnings
warnings.filterwarnings("ignore")
d = pd.read_csv("daily_sales.csv", parse_dates=["date"]).set_index("date").asfreq("D")
y = d.units_sold
print(y.shape, y.index.min().date(), "to", y.index.max().date(), "| missing days:",
y.isna().sum())
print(y.resample("YE").sum().to_dict())
print("mean units by weekday:", y.groupby(y.index.day_name()).mean().round(0).to_dict())
```

OUTPUT

```
(1461,) 2022-01-01 to 2025-12-31 | missing days: 0
{Timestamp('2022-12-31 00:00:00'): 85031, Timestamp('2023-12-31 00:00:00'): 100536,
Timestamp('2024-12-31 00:00:00'): 116798, Timestamp('2025-12-31 00:00:00'): 132920}
mean units by weekday: {'Friday': 289.0, 'Monday': 288.0, 'Saturday': 320.0,
'Sunday': 323.0, 'Thursday': 288.0, 'Tuesday': 289.0, 'Wednesday': 287.0}
```

Components of a time series



STL decomposition of daily units into trend, weekly seasonality and remainder.

Component	Meaning	Example
Trend	Long-term direction	Sales growing about 15% a year
Seasonality	Repeating pattern with a fixed period	Weekends higher; December peak; Diwali
Cycle	Longer, irregular rises and falls	Economic cycles
Remainder / noise	What is left	Random variation, one-off events

Python

```
strength_trend = max(0, 1 - res.resid.var() / (res.trend + res.resid).var())
strength_season = max(0, 1 - res.resid.var() / (res.seasonal + res.resid).var())
print(f"strength of trend {strength_trend:.2f} | strength of weekly seasonality
{strength_season:.2f} (0 = none, 1 = very strong)")
```

OUTPUT

```
strength of trend 0.95 | strength of weekly seasonality 0.64 (0 = none, 1 = very
strong)
```

Stationarity and autocorrelation

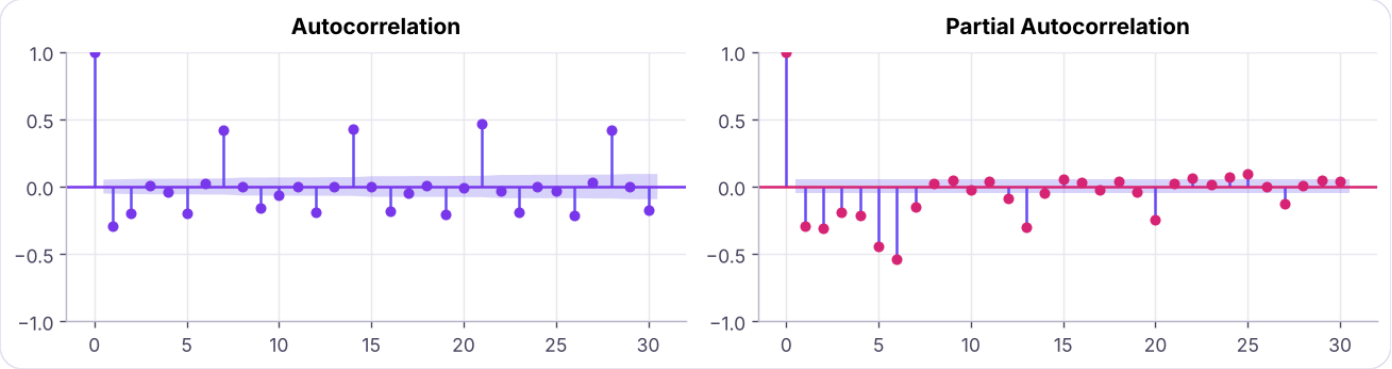
Many classical models (ARIMA) need a **stationary** series: constant mean and variance over time. Trends and seasonality break that; **differencing** removes them.

Python

```
from statsmodels.tsa.stattools import adfuller
for name, s in [("original", y), ("first difference", y.diff()), ("first + weekly
difference", y.diff().diff(7))]:
    stat, p = adfuller(s.dropna())[:2]
    print(f"{name:26s} ADF p-value {p:.4f} → {'stationary' if p < 0.05 else 'NOT
stationary'}")
```

OUTPUT

```
original                   ADF p-value 0.4455 → NOT stationary
first difference           ADF p-value 0.0000 → stationary
first + weekly difference  ADF p-value 0.0000 → stationary
```



Autocorrelation (ACF) and partial autocorrelation (PACF) of the differenced series. Spikes at lags 7, 14, 21 reveal weekly seasonality.

ACF shows correlation with past values at each lag; **PACF** shows the direct effect of each lag after removing shorter ones. They guide the AR and MA orders of ARIMA models.

Train/test split in time

Python

```
h = 90                                                  # forecast horizon: the last 90
days
train, test = y.iloc[:-h], y.iloc[-h:]
print(f"train {train.index.min().date()}–{train.index.max().date()} ({len(train)} days) |
test {test.index.min().date()}–{test.index.max().date()}")
def metrics(actual, fc):
    e = actual - fc
    return {"MAE": round(float(e.abs().mean()), 1), "RMSE": round(float((e ** 2).mean() **
.5), 1), "MAPE_%": round(float(100 * (e.abs() / actual).mean()), 2),
            "WAPE_%": round(float(100 * e.abs().sum() / actual.sum()), 2), "bias_%":
round(float(100 * -e.sum() / actual.sum()), 2)}      # bias > 0 = over-forecast
results, forecasts = {}, {}
```

OUTPUT

```
train 2022-01-01–2025-10-02 (1371 days) | test 2025-10-03–2025-12-31
```

Never shuffle time series: always train on the past and test on the future.

Metric	Use
MAE / RMSE	Error in units; RMSE punishes big misses
MAPE	Percentage error; breaks when actuals are near zero
WAPE	Total absolute error ÷ total actuals; robust, preferred in retail
Bias	Systematic over- or under-forecasting (stock-outs vs excess inventory)

Baselines

Python

```
forecasts["naive"] = pd.Series(train.iloc[-1], index=test.index)
forecasts["seasonal naive (last week)"] = pd.Series(np.tile(train.iloc[-7:].values, h // 7
+ 1)[:h], index=test.index)
forecasts["moving average (28 days)"] = pd.Series(train.iloc[-28:].mean(),
index=test.index)
for k, f in forecasts.items(): results[k] = metrics(test, f)
print(pd.DataFrame(results).T)
```

OUTPUT

	MAE	RMSE	MAPE_%	WAPE_%	bias_%
naive	19.1	22.4	5.36	5.31	1.79
seasonal naive (last week)	14.9	19.1	4.23	4.14	2.35
moving average (28 days)	20.9	23.9	5.92	5.81	2.96

The **seasonal naive** forecast ("same as last week") is the baseline every serious model must beat.

Exponential smoothing (Holt-Winters / ETS)

ETS forecasts are weighted averages of past observations, with weights decaying exponentially, plus components for **level, trend and seasonality**.

Python

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing
ets = ExponentialSmoothing(train, trend="add", damped_trend=True, seasonal="mul",
seasonal_periods=7).fit(optimized=True)
forecasts["Holt-Winters (ETS)"] = ets.forecast(h); results["Holt-Winters (ETS)"] =
metrics(test, forecasts["Holt-Winters (ETS)"])
print({k: round(float(v), 3) for k, v in ets.params.items() if k in ("smoothing_level",
"smoothing_trend", "smoothing_seasonal", "damping_trend")})
print(results["Holt-Winters (ETS)"])
```

OUTPUT

```
{'smoothing_level': 0.054, 'smoothing_trend': 0.054, 'smoothing_seasonal': 0.036,
'damping_trend': 0.989}
{'MAE': 10.4, 'RMSE': 12.8, 'MAPE_%': 2.92, 'WAPE_%': 2.88, 'bias_%': 0.34}
```

SARIMA

ARIMA(p, d, q) combines autoregression (p past values), differencing (d) and moving average of past errors (q). **SARIMA** adds seasonal terms (P, D, Q, s).

Python

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
sar = SARIMAX(train, order=(1, 1, 1), seasonal_order=(1, 1, 1, 7)).fit(disp=False)
fc = sar.get_forecast(h); forecasts["SARIMA(1,1,1)(1,1,1,7)"] = fc.predicted_mean; ci =
fc.conf_int(alpha=0.05)
results["SARIMA(1,1,1)(1,1,1,7)"] = metrics(test, forecasts["SARIMA(1,1,1)(1,1,1,7)"])
inside = ((test ⩾ ci.iloc[:, 0]) & (test ⩽ ci.iloc[:, 1])).mean()
print(f"AIC {sar.aic:.0f} | {results['SARIMA(1,1,1)(1,1,1,7)']} | 95% interval covers
{inside:.0%} of test days")
```

OUTPUT

```
AIC 11298 | {'MAE': 16.9, 'RMSE': 20.2, 'MAPE_%': 4.82, 'WAPE_%': 4.69, 'bias_%':
4.06} | 95% interval covers 96% of test days
```

Use `pmdarima.auto_arima` or `statsforecast.AutoARIMA` to search orders automatically by AIC.

Machine learning with lag features

Turn forecasting into regression: predict today's value from **lags** (same day last week), **rolling statistics** and **calendar** features.

Python

```
import lightgbm as lgb
def make_features(s):
    f = pd.DataFrame(index=s.index)
    f["dow"] = s.index.dayofweek; f["month"] = s.index.month; f["dayofyear"] =
s.index.dayofyear; f["t"] = np.arange(len(s))
    for lag in (7, 14, 21, 28, 364): f[f"lag_{lag}"] = s.shift(lag)
    f["roll_mean_28_lag7"] = s.shift(7).rolling(28).mean(); f["roll_std_28_lag7"] =
s.shift(7).rolling(28).std()
    return f
F = make_features(y)
Xtr, ytr = F.loc[train.index].dropna(), train.loc[F.loc[train.index].dropna().index]
model = lgb.LGBMRegressor(n_estimators=600, learning_rate=0.03, num_leaves=15,
min_child_samples=20, verbose=-1, random_state=0, linear_tree=True).fit(Xtr, ytr)
```

Lags are at least 7 days, so the model can forecast up to a week ahead without knowing the in-between values. For a 90-day horizon we forecast **recursively**: predict a week, feed predictions back as lags, repeat.

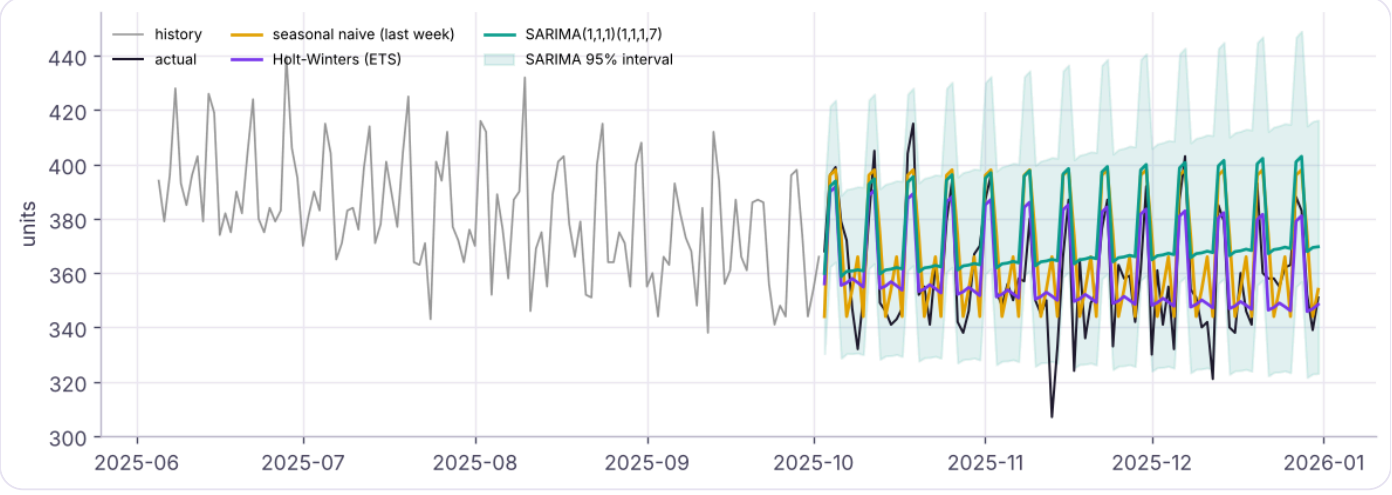
Python

```
hist = train.copy()
for day in test.index:
    s = pd.concat([hist, pd.Series([np.nan], index=[day])])
    hist.loc[day] = model.predict(make_features(s).iloc[[-1]])[0]
forecasts["LightGBM (lags, recursive)"] = hist.loc[test.index]; results["LightGBM (lags,
recursive)"] = metrics(test, forecasts["LightGBM (lags, recursive)"])
print(pd.DataFrame(results).T.sort_values("WAPE_%"))
```

OUTPUT

	MAE	RMSE	MAPE_%	WAPE_%	bias_%
Holt-Winters (ETS)	10.4	12.8	2.92	2.88	0.34
seasonal naive (last week)	14.9	19.1	4.23	4.14	2.35
SARIMA(1,1,1)(1,1,1,7)	16.9	20.2	4.82	4.69	4.06
naive	19.1	22.4	5.36	5.31	1.79
moving average (28 days)	20.9	23.9	5.92	5.81	2.96
LightGBM (lags, recursive)	21.1	27.4	6.01	5.86	4.53

Holt-Winters wins clearly here (WAPE under 3%, almost no bias), while this LightGBM set-up does worse than the seasonal naive baseline. Two reasons: errors **accumulate** when 90 days are forecast recursively from the model's own predictions, and with a single, smooth series there is little for a tree model to learn that ETS does not already capture. ML models shine with **many related series** and **external drivers** (prices, promotions, weather, holidays). The lesson: always benchmark against simple models.



The last 120 days of training data, the 90-day test period and three forecasts.

Backtesting

One holdout can be lucky. **Rolling-origin backtests** repeat the exercise from several start dates.

Python

```
origins = [pd.Timestamp("2025-03-31"), pd.Timestamp("2025-06-30"), pd.Timestamp("2025-09-30")]
rows = []
for o in origins:
    tr, te = y.loc[:o], y.loc[o + pd.Timedelta(days=1): o + pd.Timedelta(days=28)]
    snaive = pd.Series(np.tile(tr.iloc[-7:].values, 5)[:len(te)], index=te.index)
    ets_o = ExponentialSmoothing(tr, trend="add", damped_trend=True, seasonal="mul", seasonal_periods=7).fit().forecast(len(te))
    rows.append({"origin": o.date(), "seasonal naive WAPE": metrics(te, snaive)["WAPE_%"], "ETS WAPE": metrics(te, ets_o)["WAPE_%"]})
bt = pd.DataFrame(rows); print(bt.to_string(index=False)); print("mean:", bt.drop(columns="origin").mean().round(2).to_dict())
```

OUTPUT

```
    origin  seasonal naive WAPE  ETS WAPE
2025-03-31                 4.44      3.21
2025-06-30                 4.04      4.23
2025-09-30                 3.08      2.92
mean: {'seasonal naive WAPE': 3.85, 'ETS WAPE': 3.45}
```

Tools and practice

Tool	Strength
statsmodels	ETS, SARIMA(X), decomposition, diagnostics
statsforecast (Nixtla)	Very fast AutoARIMA/AutoETS for thousands of series
Prophet	Trend + seasonality + holidays with intuitive settings
LightGBM / XGBoost with lags	Many related series, external drivers (price, promotions, weather)
Deep learning (N-BEATS, TFT) and foundation models (TimeGPT, Chronos, TimesFM)	Large collections of series, zero-shot forecasts
Excel / Power BI forecast	Quick ETS forecasts for business users

Practical tips: forecast at the level decisions are made (store × product × week), add known future drivers (holidays, promotions, prices), forecast intervals not just points, monitor bias, and combine forecasts (an average of ETS and ML often beats both).

Summary

KEY TAKEAWAYS

- Decompose series into trend, seasonality and remainder (STL); difference to make them stationary (ADF test); read ACF/PACF.
- Split by time; evaluate with MAE, RMSE, MAPE, WAPE and bias; always beat the seasonal naive baseline.
- Holt-Winters (ETS) models level, trend and seasonality; SARIMA models autocorrelation and seasonal differences, with prediction intervals.
- Gradient boosting with lag, rolling and calendar features is a strong approach, especially with many series and external drivers; forecast recursively for long horizons.
- Backtest from several origins and choose by business metrics.

Quick quiz

6 questions · tap an answer to check it

1. Which baseline must every daily retail forecast beat?

A The overall mean

B Seasonal naive: the same day last week

C A random forecast

D Zero

Answer: B. It captures weekly seasonality for free.

2. What does differencing do?

A Removes trend (and with a seasonal lag, seasonality) to make the series stationary

B Smooths noise

C Adds seasonality

D Scales values

Answer: A. $y_t - y_{(t-1)}$.

3. Why is WAPE often preferred to MAPE?

A It does not explode when individual actuals are near zero and weights by volume

B It is always smaller

C It ignores errors

D It is required by statsmodels

Answer: A. Total absolute error / total actuals.

4. In SARIMA(1,1,1)(1,1,1,7), what is 7?

A The seasonal period (weekly seasonality on daily data)

B The number of lags

C The horizon

D The number of models

Answer: A. $s = 7$.

5. **Why shouldn't you shuffle data when validating forecasts?**

- **A** It would train on the future and give over-optimistic results
- **B** It is slower
- **C** It changes the target
- **D** It breaks pandas

Answer: A. Always validate forward in time.

6. **A lag-7 feature allows a model to forecast how many days ahead directly?**

- **A** Up to 7 days
- **B** 1 day only
- **C** 90 days
- **D** Any horizon

Answer: A. Beyond that, forecast recursively or use direct models.

EXERCISE

Practice

(1) Aggregate to weekly totals and forecast 12 weeks with Holt-Winters (seasonal period 52 needs 2+ years of data). (2) Add a Diwali/holiday flag feature to the LightGBM model. (3) Average the ETS and SARIMA forecasts: does the WAPE improve?

Show solution (worked answer)

Python

```
wk = y.resample("W-SUN").sum().iloc[1:-1]
ewk = ExponentialSmoothing(wk.iloc[:-12], trend="add", seasonal="add",
seasonal_periods=52).fit().forecast(12)
print(metrics(wk.iloc[-12:], ewk))
combo = (forecasts["Holt-Winters (ETS)"] + forecasts["SARIMA(1,1,1)(1,1,1,7)"]) / 2;
print(metrics(test, combo))
diwali = pd.to_datetime(["2022-10-24", "2023-11-12", "2024-11-01", "2025-10-20"])
F["festive"] = [int(any(abs((d - x).days) ⩽ 3 for x in diwali)) for d in F.index]
```

INTERVIEW CORNER

Questions you may be asked

Q: How would you forecast sales for 5,000 products in 200 stores? Hierarchical, automated approach: a global gradient-boosting model with lags, calendar, price and promotion features (or fast AutoETS per series), backtesting by store-product, reconciliation across levels, and monitoring of bias and accuracy.

Q: What is stationarity and why does it matter? Constant statistical properties over time; ARIMA-type models assume it, so we difference or transform the series first.

Q: How do you evaluate a forecasting model? Time-ordered holdout or rolling-origin backtests with MAE/WAPE and bias, comparison with a seasonal naive baseline, and checking prediction-interval coverage.

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.